FUSION X HACKATHON

TEAM_NAME: IRON MINDS
TEAM NUMBER: FXH26-P-BDDS-020

GOVSHIELD - PROJECT REPORT
"Fraud Detection for Ayushman Bharat Healthcare Scheme using DataScience and AI"

1. INTRODUCTION

Healthcare Fraud is a growing issue impacting large-scale Government Insurance programs such as Ayushman Bharat in India. Healthcare providers have made fraudulent medical claim submissions, overbilling and submitting double billing claims as well as providing fraudulent billing information about patients, creating a substantial loss of funds to the Government's operating budget for providing healthcare services.

As the volume of Healthcare Claims submitted each day grows the Manual Verification of the claims will become inefficient, time-consuming and to a degree subject to Human Error, creating a clear need for an Automated Intelligent System with the ability to identify suspicious medical insurance claims in real-time.

GovShield is an Artificial Intelligence Powered Fraud Detection System that reviews Healthcare Insurance claims for potentially fraudulent activity by applying advances in Artificial Intelligence, Data Analytics, Machine Learning, Explainable AI and the principles of Data Science; and trains the Machine Learning Models with different Technology Stacks.

2. PROBLEM STATEMENT

The existing model for health insurance claims faces numerous problems, including the following:

- Billing for services that are a fraud
- Billing of the same services more than once
- Billing for services that were not provided as part of the patient's diagnosis
- Billing for services that were billed at a higher rate than they were actually provided at
- The long manual review process that is included in most claim's invoices

The issues mentioned above collectively create a possible source of revenue loss, detract from confidence in government health care programs, and result in an overall increase in the front-end workload for all insurers in the marketplace.

To reduce these problems, the implementation of an electronic fraud detection strategy should be developed and include:

• Efficiently analyzing large claim datasets in a timely manner

• Detecting fraud patterns that are hard to identify

• Providing an initial risk score on claims received

• Supporting the auditor with their decision-making.

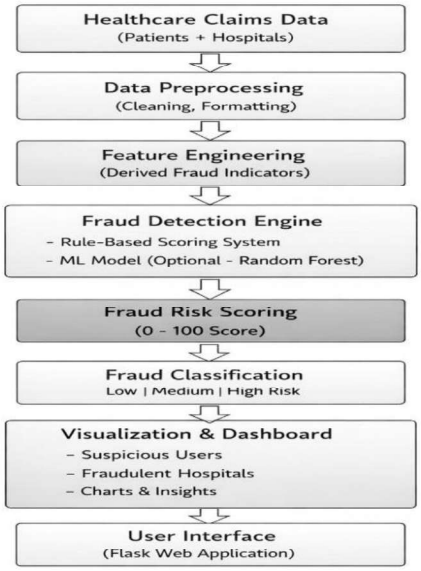
3. EXISTING SYSTEM

Fraud detection within the healthcare sector currently relies on three methods:
1) Manual methods whereby claims are verified by audit personnel. The use of manual methods is time consuming and can result in inconsistencies due to human error.
2) Rule Based methods use basic validations, which includes defining upper and lower limits for claim amounts but does not have the capability to validate the claim as a possible fraudulent transaction.
3) Standard Reporting Dashboards are primarily used to provide an overall indication of trends and performance metrics but do not provide any predictive information about possible fraud.

Limitations of Current Systems
1) No ability to provide real-time predictions.
2) Do not use machine learning or adaptive techniques.
3) Are not capable of identifying multi-dimensional episodes of fraud.
4) Do not have satisfactory scalability to accommodate larger data sets.
The design of the GovShield Fraud Detection System is a fully developed A.I based globally; its aim is at developing an end-to-end Fraud Detection and Investigation system.

Architecture Flow:

Data Ingestion → Data Preprocessing → Feature Engineering → Machine Learning Models → Fraud Risk Scoring → Interactive Dashboard → User Investigation Tools



Technologies Used:

- Python
- Pandas, NumPy
- Scikit-learn (Random Forest, Gradient Boosting)
- Matplotlib, Seaborn
- Flask / Backend API (if used)
- HTML, CSS, JavaScript
- Chart.js (for interactive visualizations)

System Modules:

Module to Create Datasets

• Create a Fake Health Claim Data Set

• Use Realistic Data Attributes about Real Indian Hospitals and Patients

• Generate Fraudulent Patterns to Train the Fraud Detection Model

Module for Creating Features

Creating New Features:

• inflation_ratio

• is_duplicate

• suspicious_volume

• income_mismatch

Machine Learning Algorithms

• Random Forest

• Gradient Boosting

• Predicting the Probability that There Was Fraudulent Activity and Associated Risk Score of a Claim from 0 to 100

Fraud Classification System

• FLAGGED - (High Risk 70+)

• REVIEW - (Medium Risk 40-69)

• CLEAR - (Low Risk 39 or less)

Interactive Dashboard

• Fraud Analytics in One Place

• Charts Based on Fraud Patterns

• Capability to Investigate Specific Claims

AI Explainable Module

• Information Sources to Validate Predictive Fraud Analysis and Fraud Detection Models

• Viewable Key Factors Associated with the Predictive Analysis of Fraud

Features for Users

• Provide Capability to Upload Their Own Datasets

• Allow User to Create Their Own Fraud Rules

• Allow User to Conduct "What-If" Simulations

• AI Assistance through Chat Bot

5. OBJECTIVES

GovShield has several main objectives that it attempts to achieve:

1. Detect fraud in health insurance claims with AI

2. Reduce the amount of money lost by governments in paying for Healthcare programs

3. Provide a risk score for fraud on a claim level when it happens

4. Support the use of an explainable AI for decision-making

5. Enable an interactive investigation of fraudulent claims

6. Make it possible to dynamically upload data and re-train models anytime new data becomes available

7. Create an efficient and transparent process for processing health insurance claims.

6.Conclusion

The GovShield platform shows that there are successful ways of using AI (Artificial Intelligence) to detect fraud in health care. The platform utilizes machine learning and provides users with an

interactive analytics environment that will help identify claims that are suspicious and explain how the claim was found to be fraudulent.

The GovShield platform provides more transparency, less manual work and a more efficient way to make decisions about the processing of health insurance claims. It is designed to be a fully scalable platform that can be adapted into real-world government healthcare systems.

7. Future Work

There are several ways that GovShield could add more value than it currently does by making improvements to the product.

* Integrate with real hospital/governmental databases.

* Use as a live monitor.

* Use advanced deep learning algorithms for fraud detection.

* Develop a mobile app for investigators of fraud in the field.

* Integrate with national health APIs.

* Create and use geo-fraud heatmaps via GIS technology.

* Develop a real-time anomaly detection streaming pipeline.

8. TECHNOLOGY STACK SUMMARY

- Python (Core backend logic)
- Scikit-learn (Machine Learning)
- Pandas & NumPy (Data processing)
- Matplotlib & Seaborn (Visualization)
- HTML, CSS, JavaScript (Frontend UI)
- Chart.js (Interactive charts)
- Flask / Node.js (Backend API if used)

Website Link: https://frauddetectorapp.netlify.app/

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